Degree: BCA, Stream: BCA

Term – II, Second Semester Examination, April - 2024

Subject Code: BCAAE201

Subject Name: Discrete Structure

Full Marks:30 Duration: 1 Hour

Part - A
Attempt 5 questions
Each question carries 2 marks (2 × 5)

1. Define Deterministic finite automata 2

or

Describe the tuples of DFA. 2

2. Describe the term implication in Propositional logic. 2

or

Describe what do you mean by the language of Finite Automata. 2

3. Define Reflexive relation with example. 2

or

Define Symmetric relation with example. 2

4. If {1,2,3,4,5,6,7} be universal set and A= {4,3,2,1}, B={2,4,6} then find (AUB)'. 2

or

Find the elements of the set {x: x ∈ N and x=n*n where 1<n<5}. 2

5. On the set A={1,3,5,7} a relation R={(1,3),(3,5),(5,3),(5,7)}.Find what kind of relation is R? 2

or

On the set A={1,3,5,7} a relation R={(1,3),(3,5),(5,3),(5,7)}.Find inverse of R? 2

Part - B,
Attempt 2 questions
Each question carries 5 marks (5 × 2)

6. Design a DFA that accepts all strings of 0, 1 divisible by 3. 5

or

Design a DFA that accepts all strings of language ab^nc where n>=1. 5

7. Let R be a relation in the set of integers Z defined by xRy if (x-y) is divisible by 5 for all x,y belonging to Z.Prove that R is an equivalence relation. 5

or

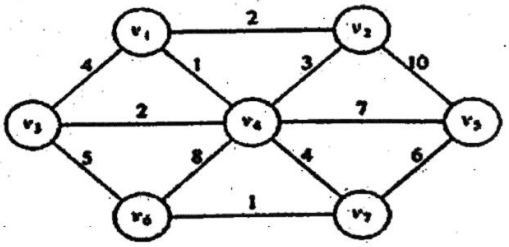
Let R be a relation in the set of integers Z defined by xRy if ((x*x)+(y*y)) is a multiple of 2 for all x,y belonging to Z. Prove that R is an equivalence relation. 5

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Using Kruskal's Algorithm find the minimal spanning tree and it's total weight from the following graph 10



or

Using Prim's Algorithm find the minimal spanning tree and it's total weight from the following graph 10

